WhatsAssist — Full Build Plan

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Week-by-Week Implementation Map

Full Tech Stack Being Built

Layer	Technology	Used In Week
Frontend	React 18 + Vite + Tailwind CSS	W5, W6, W7, W9
Backend	Node.js + Express	W5, W6, W8
Database	SQLite (sql.js - pure JS, no Python needed)	W5, W6
Auth	JWT + bcrypt (cost=12)	W5
Webhook	Meta WhatsApp Cloud API	W6, W8
AI / RAG	Gemini Flash API + vector search	W6, W8
Testing	Jest + Supertest	W10
Security	Helmet + express-validator + rate-limit	W10

File Structure (Complete)

```
New Project Final Year/
├── backend/
│   ├── src/
│   │   ├── config/
│   │   │   └── database.js          ← SQLite init (sql.js)
│   │   ├── controllers/
│   │   │   ├── authController.js    ← W5: signup, login, getMe
│   │   │   ├── webhookController.js ← W6: WhatsApp webhook handler
│   │   │   ├── ragController.js     ← W6/W8: RAG pipeline
│   │   │   ├── chatController.js    ← W6/W9: chat sessions
│   │   │   ├── knowledgeController.js← W8: doc upload & indexing
│   │   │   └── leadController.js    ← W8: lead extraction
│   │   ├── middleware/
│   │   │   ├── auth.js              ← W5: JWT middleware
│   │   │   ├── validate.js          ← W6: input validation
│   │   │   └── rateLimiter.js       ← W10: rate limiting
│   │   ├── models/
│   │   │   ├── merchant.js          ← W5
│   │   │   ├── customer.js          ← W6
│   │   │   ├── session.js           ← W6
│   │   │   ├── message.js           ← W6
│   │   │   └── lead.js              ← W8
│   │   ├── routes/
│   │   │   ├── auth.js              ← W5
│   │   │   ├── webhook.js           ← W6
│   │   │   ├── chats.js             ← W6/W9
│   │   │   ├── knowledge.js         ← W8
│   │   │   └── leads.js             ← W8
│   │   ├── services/
│   │   │   ├── geminiService.js     ← W8: Gemini AI calls
│   │   │   ├── whatsappService.js   ← W6/W8: send WA messages
│   │   │   └── ragService.js        ← W6/W8: embedding & retrieval
│   │   └── server.js
│   ├── tests/                       ← W10: Jest test files
│   ├── data/
│   ├── .env / .env.example
│   └── package.json
│
├── frontend/
│   ├── src/
│   │   ├── context/
│   │   │   └── AuthContext.jsx      ← W5
│   │   ├── pages/
```

```
│   │   │   ├── Login.jsx           ← W5
│   │   │   ├── Signup.jsx          ← W5
│   │   │   ├── Dashboard.jsx       ← W6/W7
│   │   │   ├── LiveChats.jsx       ← W9
│   │   │   ├── KnowledgeBase.jsx   ← W8/W9
│   │   │   ├── Leads.jsx           ← W8/W9
│   │   │   └── Settings.jsx        ← W8/W9
│   │   ├── components/
│   │   │   ├── Sidebar.jsx         ← W9
│   │   │   ├── ChatWindow.jsx      ← W9
│   │   │   ├── AnalyticsCard.jsx   ← W9
│   │   │   └── Toast.jsx           ← W9
│   │   ├── services/
│   │   │   └── api.js              ← W5
│   │   ├── App.jsx
│   │   └── main.jsx
│   └── package.json
│
├── week X SUBMISSION.md            ← Documentation (Weeks 1-4 done)
├── README.md
└── .gitignore
```

Known Issues Fixed

> IMPORTANT
>
> `better-sqlite3` **fails on Node v24 (needs Python/C++ build tools) Fix:** Switching to `sql.js` — pure JavaScript WebAssembly SQLite, zero compilation required.

Status

Week	Status	Notes
Week 5	🔄 In Progress	Auth code written, DB fix needed
Week 6	⏳ Pending	Waiting for W11-W12 tasks
Week 7	⏳ Pending	Midterm gate
Week 8	⏳ Pending	
Week 9	⏳ Pending	
Week 10	⏳ Pending	
Week 11	❓ Awaiting	User to share
Week 12	❓ Awaiting	User to share